

# **The Future of Computational Intelligence in Surgical Decision-Making**

**To:** Office of Science and Technology Policy (OSTP)

**From:** Meghan Dierks, MD, FACS [Fellow of the American College of Surgeons]; Chief AI Officer, HHS

**Date:** March 15, 2025

**Subject:** R&D and Investment Needs to Secure the Future of Computational Intelligence in Surgical Decision-Making

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## **Current Challenges and Limitations of AI in Surgical Settings**

The operating theater represents one of the most complex decision environments in contemporary healthcare delivery - particularly in the context of trauma surgery (e.g., battlefield) and re-operative surgery. Unlike many other procedural contexts (e.g., interventional cardiology working exclusively within a vascular lumen) where workflows, instrumentation, and anatomic relationships can be fully anticipated and mapped in advance, trauma and re-operative surgery demands continuous adaptation to unexpected findings, anatomical variations, and emergent complications. The necessity for continuous adaptation to emergent conditions - responding to unexpected findings, managing uncertainty, and adjusting to evolving tissue responses or technical execution errors - represents the fundamental challenge of surgical contingency planning and robotics advancement. This complex reality creates both unprecedented opportunities and distinctive hurdles for computational intelligence in the operating theater, substantially advancing current commercial robotics products whose platforms are limited to enhanced tool manipulation.

Large Language Models (LLMs) trained on available clinical data offer promising capabilities for general medical knowledge retrieval and pattern recognition, but they face significant

limitations in the surgical procedural context. The statistical nature of these models creates blind spots precisely where surgical decision-making requires the greatest precision:

- Clinical training data inherently underrepresents complications and rare events, skewing model understanding toward common, successful cases. An extensive review of operative reports and surgical morbidity and mortality case reviews by this author over the past 35 years confirms the parsimony and under-representation of events and observations in this corpus of training data, primarily driven by medical-legal concerns.
- Surgical complications emerge from complex causal chains with temporal dynamics that current probabilistic models struggle to capture faithfully.
- The tacit knowledge that experienced surgeons employ when navigating unexpected scenarios remains inadequately documented and formalized in the published literature that forms LLM training data.

These limitations create a fundamental mismatch between the statistical reasoning of current AI systems and the causal, constraint-based reasoning required for effective surgical contingency planning.

## **The Computational Requirements for Embedded Real-Time Surgical Intelligence and the Case for Symbolic Reasoning as Critical Augmentation**

AI technology required to support real-time surgical decision-making or embedded intelligence in surgical robotics requires specialized computational architecture designed for both performance and reliability:

- **Latency constraints** demand response times under 200ms for time-critical guidance
- **Contextual awareness** necessitates extensive context windows (50K-100K tokens) to maintain the full procedural history
- **Parallel processing** of multiple potential contingency pathways simultaneously
- **Uncertainty quantification** with explicit representation of confidence in different intervention options, requiring specialized inference optimization

- **Specialized management of complexity** with dynamic model pruning based on procedure type and resource allocation algorithms that prioritize time-sensitive inference during critical surgical phases (so-called critical path prioritization).

However, the core challenge extends beyond raw computational power to the fundamental reasoning architecture needed for surgical intelligence. Pure neural approaches to surgical intelligence (i.e., based exclusively on LLM/foundational models with fine-tuning on domain-specific data) face inherent limitations that will require hybrid approaches that symbolic reasoning systems. Symbolic systems can formally encode anatomical relationships, physiological parameters, procedural constraints, and human behavioral variations (e.g., models of risk tolerance) that must be satisfied regardless of statistical patterns. This provides essential guardrails for AI-generated contingency plans. Formal temporal logic frameworks can explicitly represent the evolution of surgical scenarios over time, including critical intervention windows and causal relationships between actions and outcomes. For example, when managing complex intra-abdominal trauma, multiple critical factors, including progressive blood loss, deteriorating metabolic parameters, declining core temperature, limited resource availability, and competing demands for resources in a mass casualty scenario, may necessitate the rapid formulation and execution of a sequentially staged approach. Such damage control strategies might involve temporary abdominal packing and closure, strategic pausing to assess survivability and potential disability outcomes, followed by definitive repair during a subsequent procedure when the patient has been physiologically stabilized.

Symbolic methods enable formal verification of proposed interventions against explicit safety criteria, providing necessary assurance for high-stakes surgical decisions. The most promising path forward for innovation and R&D requires a hybrid neuro-symbolic architectures that combine the pattern recognition strengths of LLMs with the precision and constraint satisfaction capabilities of symbolic systems.

## **Cross-Domain Inspiration: Learning from Other Sectors**

Several non-medical domains have pioneered in embedded intelligence for high-stakes, real-time decision support; these may offer insights and blueprints for advancement in surgical applications. The aerospace industry has developed rigorous frameworks for human-AI collaboration in contexts where failures are catastrophic. These systems blend probabilistic reasoning with hard constraints and feature explicit anomaly detection that could be adapted to identify developing surgical complications. Chemical processing and other high-risk industrial facilities employ fault tolerance systems with formal verification methods and digital twin technology that could be repurposed to provide real-time simulation of alternative surgical approaches. Self-driving systems feature commercially viable edge-computing implementations for real-time scene understanding, deploying localized uncertainty quantification algorithms and safety envelope enforcement through symbolic guardrails that operate directly on in-vehicle neural systems without requiring cloud connectivity.

Despite the transformative potential of surgical intelligence systems, a persistent barrier to advancement remains the misalignment of commercial incentives, where proven technologies from aerospace, autonomous vehicles, and industrial processing systems have not been applied to surgical contexts due to smaller markets, higher regulatory hurdles, and entrenched clinical workflows that discourage cross-domain technology transfer.

## **What is Required to Advance AI-Surgical Intelligence R&D?**

Advancing this field requires strategic research priorities across multiple dimensions, including model development, infrastructure, and validation frameworks. The following summarizes recommended priorities:

### **Technical Research Priorities**

1. Development of causal inference models specifically designed to learn surgical cause-effect relationships from limited data
2. Creation of explainable hybrid systems that provide reasoning transparency during critical decisions

3. Advancement of few-shot learning frameworks for rare complications with minimal training examples
4. Application of *safety envelope enforcement* concept in a surgical context (e.g., explicit anatomical no-go zones, computationally defined regions containing critical structures that cannot be violated regardless of what the neural network component might suggest, operating through real-time geometric constraint checking, where each proposed instrument movement is verified against formalized anatomical safety maps before execution).

### **Infrastructure Requirements**

1. Establishment of modular architecture standards allowing component-wise validation
2. Construction of comprehensive surgical knowledge graphs with formal representation of procedural knowledge
3. Development of multi-center federated learning systems to capture diverse patient populations

### **Validation Approaches**

1. Simulation-based certification protocols before clinical deployment
2. Gradual autonomy frameworks similar to autonomous vehicle levels, establishing clear progression paths for surgical AI capabilities
3. Surgical digital twins for risk-free testing of embedded intelligence

### **Investing in the Development of Gradual Autonomy Frameworks for Surgical Intelligence**

Even with advances in model development and technical infrastructure, one should anticipate substantial regulatory and adoption hurdles. The implementation of AI in surgical settings requires a methodical, stepwise approach to ensure safety and clinical efficacy. To overcome such hurdles, there should be equal R&D investment into the development of novel validation frameworks. Drawing inspiration from autonomous vehicle development, a

gradual autonomy framework for surgical intelligence provides a structured pathway for research, validation, and clinical integration.

### **Level 1: Enhanced Visualization and Awareness**

At this initial level, AI systems provide information augmentation without direct intervention recommendations:

- Real-time identification and labeling of anatomical structures
- Detection of tissue boundaries and critical structures
- Physiological monitoring with early warning indicators
- Tracking of instrument positioning relative to critical structures

The surgeon maintains complete control, with the system providing enhanced situational awareness similar to advanced radar systems in aviation.

### **Level 2: Decision Support with Surgeon Validation**

The second level introduces passive decision support capabilities:

- Suggestion of potential next steps in the procedural workflow
- Identification of deviation from planned approach
- Early warning of developing complications based on subtle indicators
- Presentation of relevant case history or literature for emerging scenarios

All suggestions require explicit surgeon approval before implementation, maintaining a clear human-in-the-loop paradigm.

### **Level 3: Procedural Assistance with Supervision**

At this level, systems can perform specific subtasks under direct supervision:

- Camera positioning and stability maintenance
- Tissue retraction assistance
- Dissection path suggestion with real-time adjustment

- Suture placement guidance with tension monitoring

The fully-trained surgeon supervises all activities and can immediately override any system action.

#### **Level 4: Autonomous Execution of Defined Subtasks**

With sufficient validation, specific well-defined subtasks may be delegated:

- Standardized suturing of predetermined tissue planes
- Precise tumor margin identification and marking
- Optimal port placement based on patient-specific anatomy
- Tissue perfusion assessment and mapping

The fully-trained initiates each subtask and maintains override capability throughout.

#### **Level 5: Contingency Management**

Advanced systems would incorporate the ability to:

- Automatically adjust to unexpected anatomical variations
- Identify developing complications and initiate appropriate responses
- Modify surgical approaches based on intraoperative findings
- Continuously recalculate risk profiles as the procedure evolves
- Incorporate meta-system events and states (e.g., mass casualty, resource reserves, competing demands for resources) to adjust case-specific procedural plans

These capabilities would be implemented with extensive safeguards and validation protocols.

A graduated approach ensures that surgical AI earns trust incrementally through demonstrated capability and reliability rather than attempting revolutionary but potentially unsafe paradigm shifts.

### **The Path Forward**

The future of surgical intelligence lies not in replacing clinical judgment but in augmenting it with systems that blend the complementary strengths of neural and symbolic approaches. By learning from advances in aerospace, financial, and autonomous systems, we can develop hybrid architectures that maintain human surgical expertise as the final authority while providing increasingly sophisticated decision support. Achieving this vision, however, requires overcoming significant economic barriers that currently impede progress. The surgical technology market, while substantial, remains fragmented and risk-averse compared to consumer technology sectors. Despite proven concepts in adjacent fields, private sector investment has been insufficient to bridge the "valley of death" between promising research and clinically viable systems—particularly for technologies requiring lengthy regulatory pathways and hospital adoption cycles. This market failure represents a classic case for targeted federal investment through agencies like the National Science Foundation. NSF funding for high-risk, high-reward surgical AI concepts that lack immediate commercial viability but promise significant patient benefit would accelerate progress in ways that market forces alone cannot achieve. By supporting pre-competitive research in surgical intelligence, particularly in domains like symbolic reasoning, formal verification, and human-AI collaboration models, NSF and other federal R&D support can catalyze an ecosystem that eventually becomes self-sustaining through later-stage private investment.

The most promising immediate direction combines the redundancy and verification systems from aerospace with the real-time decision capabilities from industrial processing systems, creating a new class of surgical intelligence that respects the unique contingency requirements of the operating theater while enhancing the surgeon's ability to navigate unexpected scenarios safely and effectively. With strategic federal support to overcome current market limitations, we can accelerate this integration and dramatically improve surgical outcomes across diverse healthcare environments worldwide.